Motorcycle Throttle-By-Wire Cruise Control

Installation Manual

- KTM 690 Enduro(R) and SMC
- Husqvarna 701 Enduro(R)
- Gas Gas ES 700 Enduro(R)

Model years from 2014 to 2024

VERIDIAN CRUISE

249788 Alberta Limited operating as

Veridian Cruise

Calgary, AB, Canada

Last Updated: August 2023

Check Website for latest version: http://veridiancruise.

Pack List:

- [x] Control Module with KTM 690 Firmware
- [x] Wiring Harness KTM 690
- [x] 8" Zip Ties x 8
- [x] Sticker!
- [x] Wire Tap
- [x] Option A: Triple Switch 50cm
- [ ] Option B: Soft Touch Double Switch 50cm
- [ ] Option C: Micro Toggle Switch
  - Dress Nut
  - Mounting Plate

Packing Slip

Veridian Cruise
1212 34 Ave SE
Calgary, AB T2G 1V7

VERIDIAN CRUISE

Ship To: JIM ROAL
7646 HARMONY PEAK ST
LAS VEGAS, NV 89166-5204 US

Order # 5134
Date 6/18/2025
User jimroal@gmail.com
Ship Date

Item	Description	Qty
1070	KTM 690 / 701 Enduro Cruise Control EURO4 Cruise Kit - Triple Switch (50cm) Switch Style: Triple Switch (50cm)	1
xxx7	KTM OFF ROAD/ABS/BAD FUEL Software	1

2023 Husqvarna 701 Enduro US model with white diagnostic connector. I also want the ABS add-on feature.

Happy Trails!

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Warranty

The manufacturer warrants this product to be free from defects in workmanship and materials, under normal use and conditions, for a period of two (2) years for the original invoice date. The manufacturer agrees, at its option during the warranty period, to repair and defect in material or workmanship or to furnish a repaired or refurbished product of equal value in exchange without charge (except for a fee for shipping, handling, packing, return postage, and insurance which will be incurred by the customer). Such repair or replacement is subject to verification of the defect or malfunction and proof of purchase as confirmed by showing the model number on original dated sales receipt.

Background

Veridian Cruise is owned and operated by a group of engineer friends who love to tinker and make our older bikes better. Our aspirations are to share some of the creations with the public in a safe and tested manor. We are committed to providing quality, reliable products.

Throttle by wire systems have proven to be very robust however they are sensitive electronics and must be treated as such. We think riders deserve to know the facts: corrosion, water ingress and electrical noise are serious potential threats to vehicles, and we do not take them lightly.

Our systems are designed without too many bells and whistles intentionally. A cruise control module should only have limited functionality. This inherently reduces the risk of any potential software bug. Where possible we use isolated interfaces and controllers to reduce the risk of noise. Circuit boards are designed for automotive use and meet Automotive AEC-100 Grade 2, which permits operation between -40° to 105°C.

WARNING

IN THE EVENT OF ANY UNEXPECTED BEHAVIOUR RELATING TO THE THROTTLE, EITHER WITH THE CRUISE CONTROL TURNED ON OR NOT, PULL THE CLUTCH IN TO DISCONNECT DRIVE FROM THE REAR WHEEL AND USE THE 'KILL' OR ENGINE STOP SWITCH TO TURN THE ENGINE OFF.

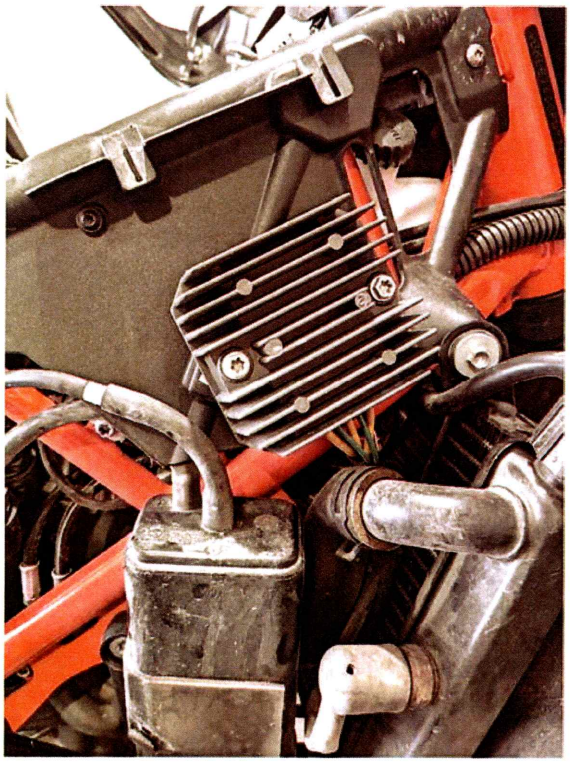
Testing has resulted in programming to deliver safe, reliable operation on this motorcycle. It is essential that you install the cruise control in accordance with the advice in the installation instructions precisely so that electrical interference does not cause the vehicle or cruise control to behave erratically or be rendered inoperative.

Installation

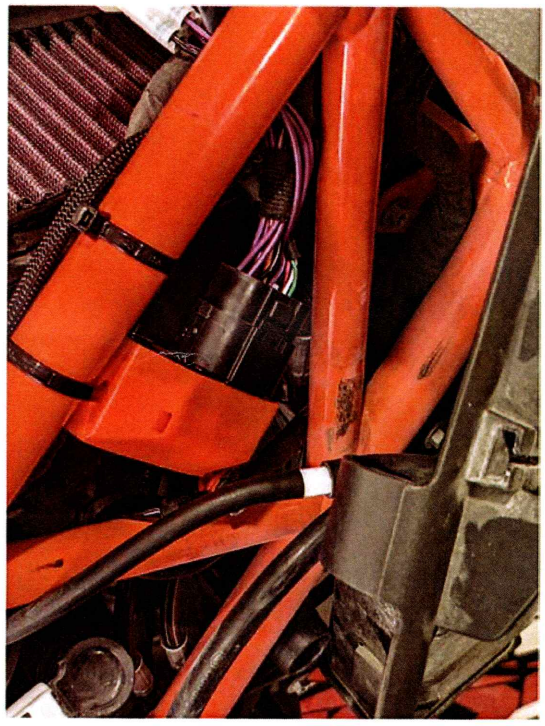
NOTE: - It is recommended that the fuel tank is less than 1/4 full before attempting to fit the cruise control. The fuel tank must be lifted for most installations and can be very heavy when full of fuel.

1. Remove Seat and Battery.
2. Remove Battery base. (3x 8mm bolts)
3. Remove air filter scope. (2x 8mm bolts)
4. Remove right side tank plastics. EURO5 bikes will be LEFT side. (2x 8mm bolts)
5. Unbolt fuse box. (3x 8mm bolts)
6. Install the cruise controller into the frame area as shown. EURO5 and bikes equipped with Tile/Lean sensor may find more room on the Left side of the steering tube.
7. Locate, and disconnect the grip sensor plugs. This is a translucent white six-way connector located just above the airbox near the steering tube. Connect the matching T-harness plugs.
8. Connect the CANBUS connector to the diagnostic port under the rear seat. Use the remaining zip ties to secure the cable to the frame. It is critical not to put this cable anywhere it could get pinched, rubbed, or caught in any moving parts.
9. Connect Purple Brake input wire to the White with Green wire located under the fusebox. The same plug for the rear brakes lamp. Use the included T-Tap. Dielectric grease is recommended for water tightness. It is preferred to connect before the 3 pin taillamp connector in the even the connector gets corroded. Once complete this input can be tested by pulling the brake lever. The controller will respond with flashing red and yellow LED. *This connection is optional for the 2021+ year and will provide rear brake input.*

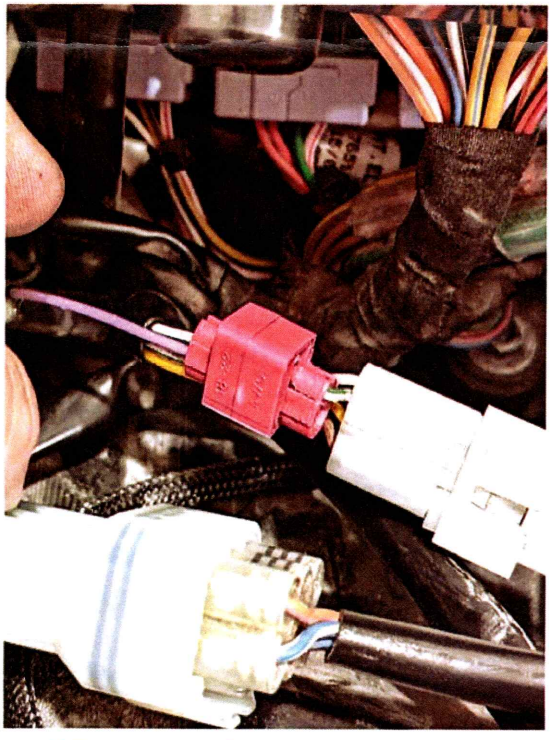
*See your bikes repair manual. Contact us if you need help locating a copy of your repair manual. Not to be confused with the owner's manual.



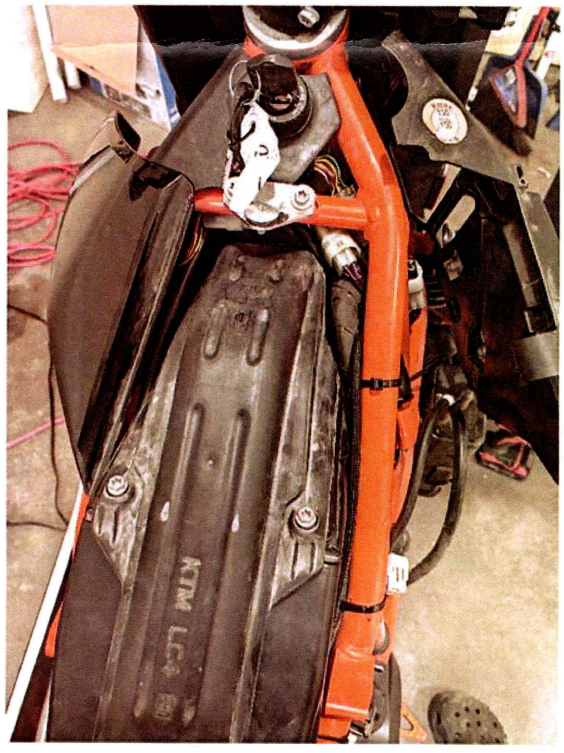
Remove Right side tank plastic



Install Area for controller



Use T-Tap on White/ Green Wire.



Final install routing

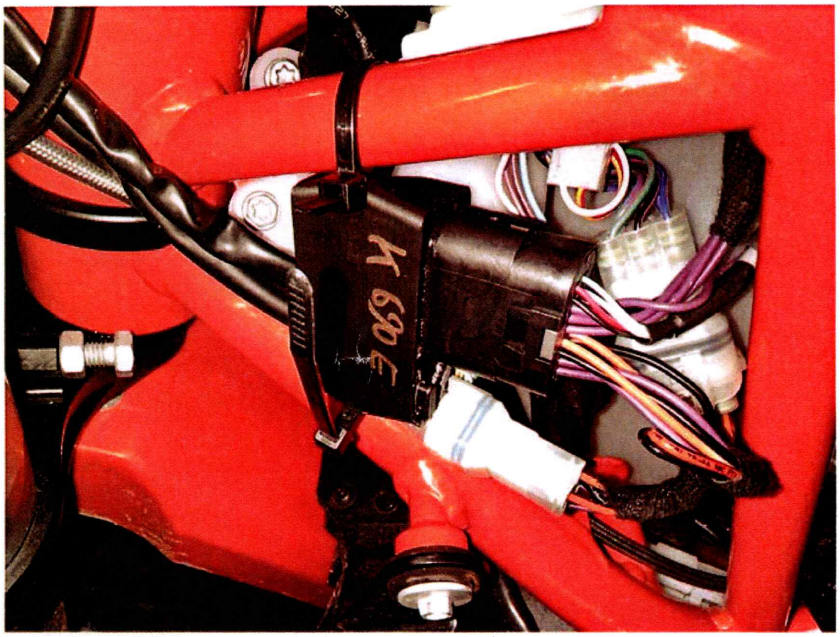


Figure 1 Mounting position for Bike with Tile Lean

Finishing up

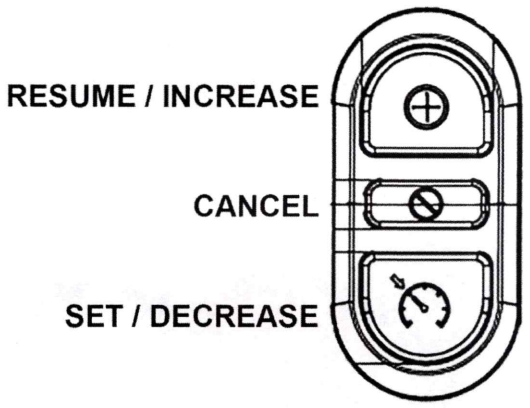
1. Re-fit plastics and airbox
2. Re-fit battery and seat.

Notes

- Check the routing of the wires and fit cable ties where necessary.
- Check carefully for moving or stationary objects that might damage any wires.

Operation

The controller has been pre-programmed for your bike; no programming is required. For advanced users and troubleshooting please see the programming section if you want to tinker.



Operation is quite standard to most vehicles. When the controller is disabled. Throttle signals are passed directly to the ECU. No processing or re-mapping is done to ensure maximum compliance with the OEM safety/ diagnostics. If the twist grip sensor fails, those signals will be passed to the ECU.

From the disabled state:

- Pressing SET, If there are no active kill conditions (See below), Cruise controller is enabled, Target speed is set at the current bike speed.
- Pressing RESume, if there are no active kill conditions, (See below), Cruise controller is enabled, Target speed is resumed from a previously set condition or current speed if none was set.
- Pressing Cancel, Does nothing in this state.

From the enabled state:

- Pressing SET, Target speed is decreased by the programmed amount. Default value is 3 KPH per press. See programming section for more detail.
- Pressing RESume, Target speed is incremented by the programmed amount. Default value is 3 KPH per press. See programming section for more detail.
- Pressing Cancel, Put the cruise controller back into Disabled state.

Kill Conditions

This cruise control module has been designed with safety first. We have programmed the device to disable cruise control if any one of the following events are met:

- Any brake lever, front or rear is pulled enough to activate the brake lights.
- Any one of the required CANBUS packets are not updated within 1000 milliseconds.
- Input from Throttle tube exceeds cruise control output.
- A difference of 15% between the front and rear wheels. This may occur on bumps and loose conditions.

Unlike our other kits, this system will **NOT** cancel cruise for the following inputs:

- Change of transmission gear.
- Clutch lever is activated.
- Engine RPM signal falls below 1000

Offroad Addon (If Purchased)

This software enables additional features in the ABS control unit.

For KTM 690 bikes, the default mode when powered on is Mode 1, ABS only on the front wheel. Indicated by a slow flashing ABS light.

Holding the ABS button for 3-5 seconds will change to mode 2, No ABS. Indicated by a steady flashing light. After power reset, bike will revert to mode 1.

To Toggle the software addon ON and OFF:

1. Ignition ON, engine OFF, throttle in closed position.
2. Press the SET button 4 times slowly.
3. Press the CANCEL button. Any other button or timeout of 60 seconds will cancel this procedure.
4. Offroad mode will now toggle on or off.

Throttle Programming

The controller has been pre-programmed for your bike; no programming is required. However, if you suspect problems with this factory setup, manual throttle setup is possible. If you get a check engine light, try this first.

1. Ignition ON, engine OFF, throttle in closed position.
2. Press the SET button 5 times slowly. Controller LED will flash Blue slowly.
3. HOLD Throttle fully. Press the CANCEL button. Any other button or timeout of 60 seconds will cancel this procedure.
4. Throttle positions are now learned.

Control LED

Red – System is not ready for engagement. CANBUS missing messages.

Yellow – System is not ready for engagement. See Kill Conditions. Connect USB cable for advanced diagnostics.

Red / Yellow Flashing – Brake is depressed.

Blue – System Ready, Cruise is ready to enable. No signal modification to throttle.

Green – Cruise is enabled at set point. Throttle signals will be modified by the controller.

USB Programming

For Advanced Users and diagnostics. A variety of conditions can be changed with the USB-C terminal. Remove the USB-C cover and connect the controller to a PC or Android phone with a Serial Terminal app.

Function (Not case Sensitive)	Arguments	Comments
DFU	None	Enter bootloader mode for firmware upgrades
Config	None	Print out the current config
Stopcode	None	Prints the last stop code from Cruise Cancel event. See Kill Conditions. Memory lost on power.
Debug	None	Start/Stop printing Debug messages. While cruise is active: 3 data fields are: cruiseSetPoint, CAN_Rear_Wheel, output_throttle. Use for custom tuning PID.
Reset	None	Reset Config
Help	None	Print Help
Wheelx	18	Adjust the wheel speed sensor factor. Change for custom applications or extreme tire changes.
SpeedINC	05	Adjust the amount to increase speed with a single press of the INC/DESC buttons. Units are KM/H
PID	120 05 00	Adjust the PID gain
tmax	50	Adjust the Maximum throttle limit. Units are 1%
TPSidle	None	Set the throttle input idle position
TPSopen	None	Set the throttle input open position
Offroad	None	Toggle offroad software addon ON and OFF

Example:

speedinc 04

The above command will set the new speed increment to 4kph.

PID 200 15 05

The above command will set the PID controller with values of P=200, I=15, D=05

Disclaimers and Disclosures

Buyer, his/her heirs, assigns, legal representatives and attorneys, knowingly and voluntarily release any and all claims, demands, or causes of action, known or unknown, which he/she may have against the seller, his/her successors, assigns, representatives and lessees arising from any injury or illness, including death or disability, incurred in connections with the installation and use of this product. Buyer assumes all responsibility for the installation and maintenance of this product. Improper installation, improper maintenance, or deliberate misuse shall also release seller, his/her successors, assigns, representatives and lessees from any and all claims, demands, or causes of action, known or unknown arising from any injury or illness, including death or disability, incurred in connections with the installation and use of this product.

1. Improper installation includes, but is not limited to:

- Failure on the part of the buyer to properly install all the supplied parts and components in accordance with the supplied instructions.
- Failure to install ALL the supplied parts and components in accordance with the supplied instructions.
- Modification of ANY parts or components.

2. Improper maintenance includes, but is not limited to:

- Failure to periodically inspect installed parts and components for tightness, wear and/or damage.
- Failure to periodically inspect and lubricate installed parts and components in accordance with the supplied instructions.
- Modification of ANY parts or components.

3. Misuse includes but is not limited to:

- Using the installed item for purposes other than which it is intended.
- Modification of ANY parts or components.

All Veridian Cruise labor and services are performed after an agreement with the vehicle's owner and the owner must accept the responsibility and terms of this liability form. The owner of any Veridian Cruise product or vehicle on which labor was performed which is sold, traded, or gifted must notify the new owner of these conditions and terms.

Veridian Cruise aftermarket products modify a vehicle for uses which exceed conditions anticipated by the vehicle manufacturer. Veridian Cruise products are not designed to meet vehicular code, federal law, state law, or any other vehicle requirement. Owners accept all responsibility for verifying the use of Veridian Cruise products in their state prior to installation and use. Veridian Cruise products have been designed for the high performance demands of off-road driving, four wheeling, and rock crawling.

Hold Harmless/Liability

Veridian Cruise is not liable nor held responsible for any damages, injury's or death that can occur from on and off-road use, the result of failure of our product's, vehicular accidents, roll-overs, installation error, or lack of maintenance. Customer/driver agrees to hold harmless Veridian Cruise and further assumes all liability for using any Veridian Cruise product(s) and in assuring that the system or product accessory is properly installed, maintained, and operating in safe conditions. The use of Veridian Cruise products varies in the degree of extremity and cannot be controlled by the vehicle or product manufacturer. Do not purchase Veridian Cruise manufactured products or components unless you are willing to accept this responsibility.

Need Help?

Send us an email. info@veridiancruise.com or check out website for the latest manuals.

These manuals are still a work in progress and we appreciate any feedback.



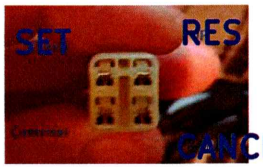
Use 22-18AWG T-Tap
Tap into White/Green wire under Fusebox to tail lamp

BRAKE_CANCEL 1 — J3 T-Tap



Euro5 ISO / DIS 19689 6-Pin Connector	
CAN_ H line	2
Ground	3
VBAT +12V	4
CAN_ L line	5
K-Line	6

Female for switch side





J5 Sumitomo HM-4P
1 RES, 1.70mm, Pink
2 SET, 1.70mm, Y/Red
3 CANCEL, 1.70mm, Purple
4 CRUISE_COMMON, 1.70mm, Grey/ Red

LENGTH: 22in

For EURO5
J8 MWTPB-06-1A-R
1
CANH 2
CAN_GND 3
CAN12V 4
CANL 5
KLine 6

CAN12V 1
CANH 2
CANL 3
4
5
CAN_GND 6
J9 Sumitomo MT-6S-2
Bike

Shielded pair

LENGTH: 30in

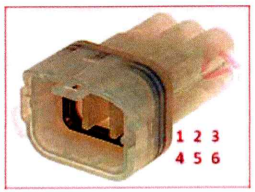


18 Pin: MX23A18SF1
Wire: 22-16AWG 1.6mm to 2.2mm sheath
Front Cap: MX23A18XF1
Socket Contact M23S05K351
Dummy Pin M120-55780

J2 MX23A18SF1

CANL	9	18	CAN12V
CANH	8	17	CAN_GND
CRUISE_COMM	7	16	RES
	6	15	CANCEL
BRAKE_CANCEL	5	14	SET
5V	4	13	5V
GND	3	12	GND
TTBOUT	2	11	TTAOUT
TTAIN	1	10	TTBIN

External Cancel will Cancel cruise on a HIGH 12V Signal.
Example brake wire.

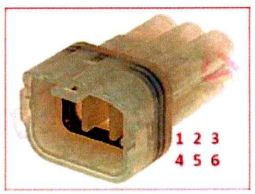


Heat Shrink or TechFlex

Colours:
Black - Ground
Red - 12V
Orange - 5V
Purple - Signals

LENGTH: 4in





J1 Sumitomo MT-6S-2 Throttle
1 TTAIN
2 GND
3 5V
4 TTBIN
5 GND
6 5V

J7 Sumitomo MT-6P-2 ECU
TTAOUT 1
GND 2
5V 3
TTBOUT 4
GND 5
5V 6

Veridian Cruise

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Title: KTM 690 Enduro/SMC 14+

Size: USLetter	Date: 2023-08-17	Rev:
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